

Mathematical Modeling & Control Strategies for Hybrid Hydrogen Systems

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Plug-in FCHEV · PEMFC Modeling · Optimal EMS

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Presentation Outline

- 01 Motivation & Problem Definition
- 02 System Architecture & Selections
- 03 Mathematical Models (Vehicle · PEMFC · Battery)
- 04 Cost Function Derivation
- 05 EMS Strategies (DP · A-ECMS · MPC · BLFS)
- 06 Status & MAT 4902E Roadmap

Why Hybrid? The Case for FCHEV

PEMFC Alone — Limitations

- ✗ Slow dynamic response to acceleration
- ✗ Cannot absorb regenerative braking energy
- ✗ Frequent load cycling → membrane degradation



FCHEV Hybrid — Solution

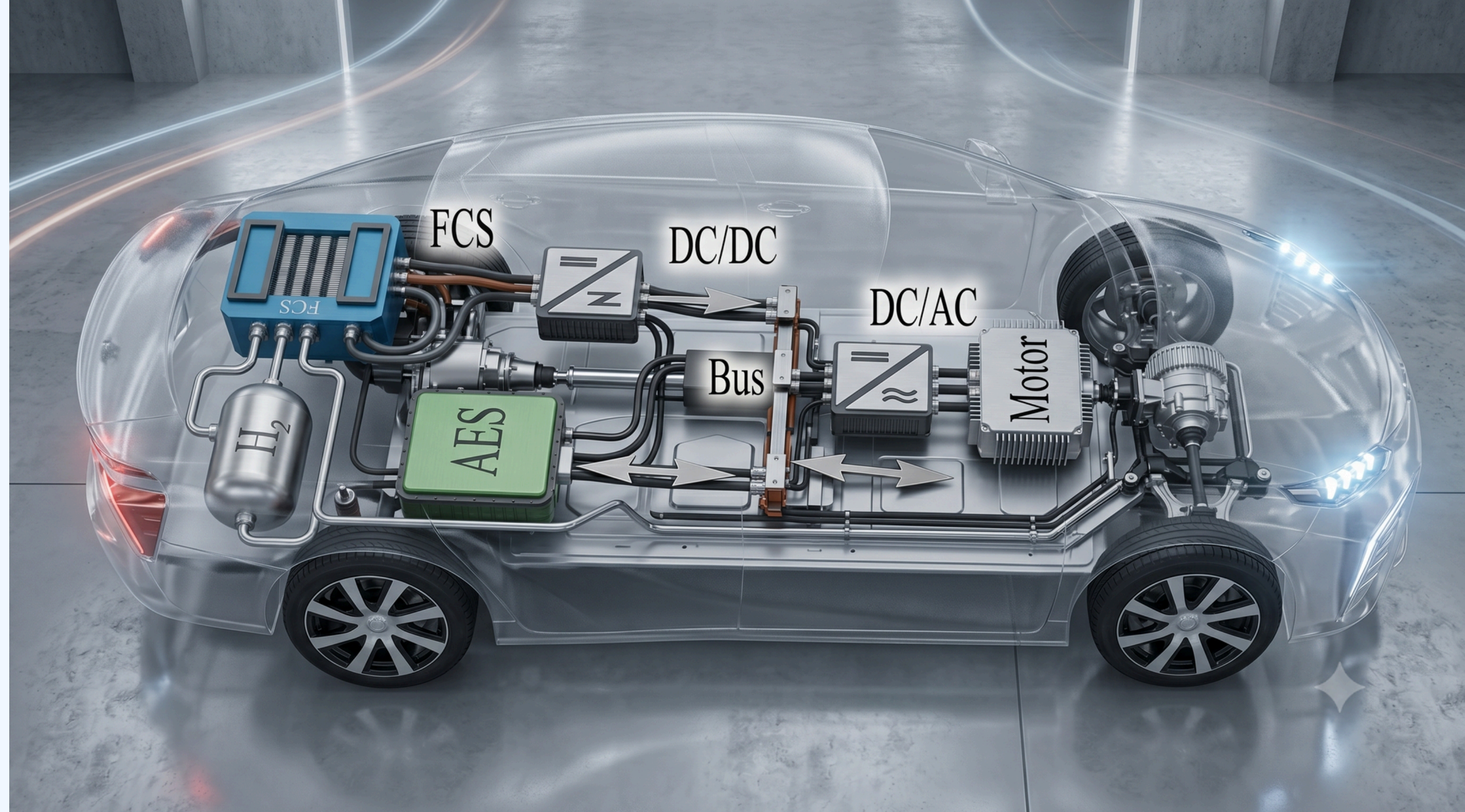
- ✓ Li-ion battery handles transient power demands
- ✓ Battery absorbs regenerative braking energy
- ✓ FC operates in efficient steady-state band

Key Question: At every instant t , how much power should the FC supply vs. the battery? → This power-split decision rule is the Energy Management Strategy (EMS).

System Architecture & Selections

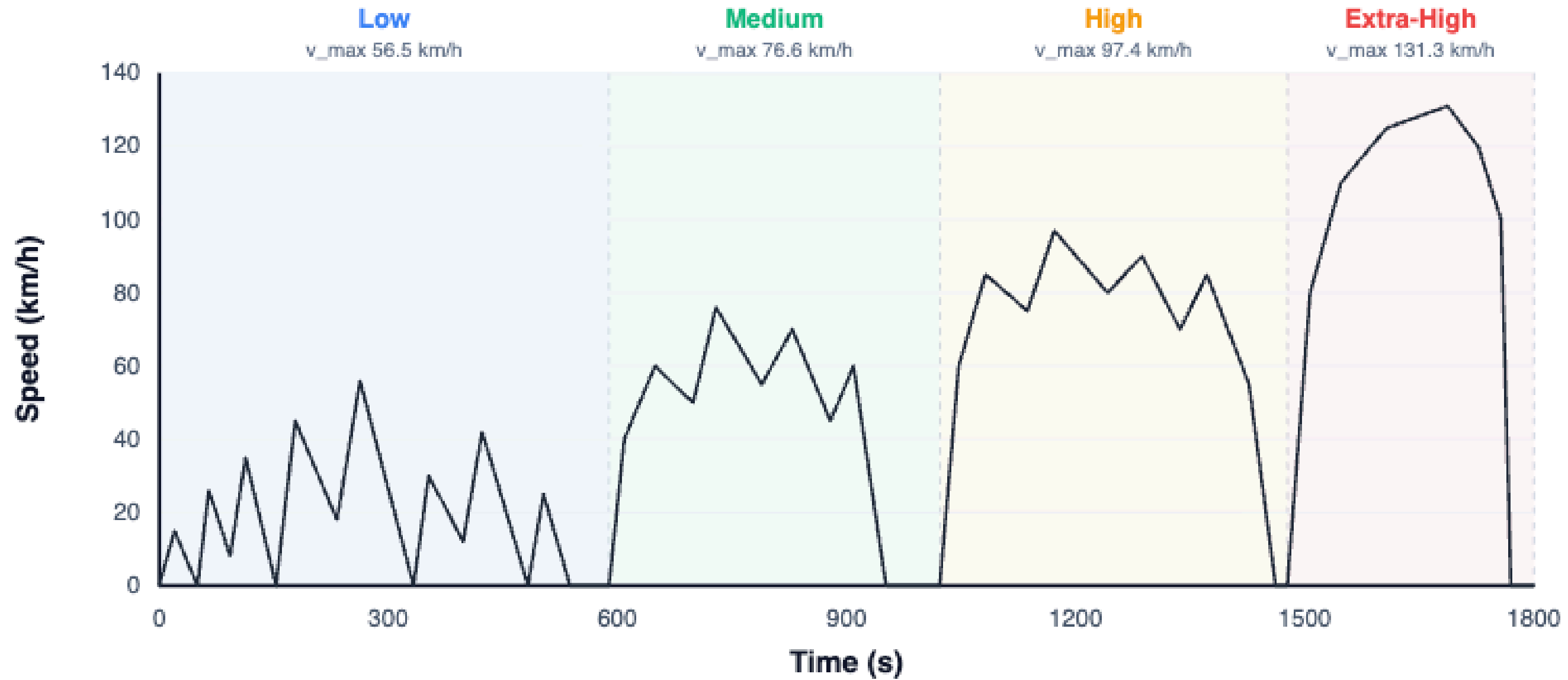
Component	Selection
Topology	T4: PEMFC → DC/DC converter → DC bus ← Battery (direct)
Reference Vehicle	2nd-Gen Toyota Mirai (2020+) m = 1925 kg, P_fc,max = 128 kW
Drive Cycle	2 × WLTP Class 3b (T = 3600 s · 46.5 km · N = 3600 decision points)
PEMFC Model	Mann–Amphlett 0-D Semi-Empirical (GSSEM)
Battery Pack	A123 ANR26650M1B LiFePO ₄ 96s10p 7.92 kWh SoC ₀ = 0.90
Battery Model	1-RC + Hysteresis Equivalent Circuit (Nejad et al. 2016)
Cost Scenario	Germany 2026: €11/kg H ₂ · €0.35/kWh electricity
EMS Strategies	DP · Adaptive ECMS · MPC · BLFS

T4 Topology



WLTP Class 3b Driving Cycle

Vehicle speed vs. time — four phases (illustrative profile)



Duration

1800 s

Distance

23.27 km

Avg speed

46.5 km/h

Max speed

131.3 km/h

Stops

~9

Models — Vehicle Dynamics & Power Balance

Longitudinal Dynamics — Wheel Power

$$P_{\text{wheel}} = [m_v a + m_v g C_r \cos \alpha + \frac{1}{2} \rho_{\text{air}} C_d A_f v^2 + m_v g \sin \alpha] v$$

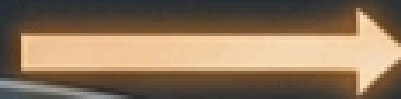
DC Bus Power Balance — defines the EMS decision variable $u(t) \equiv P_{\text{dc}}(t)$

$$P_{\text{bat}}(t) = P_{\text{load}}(t) - P_{\text{dc}}(t)$$

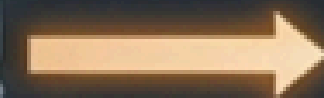
$u(t) \equiv P_{\text{dc}}(t)$ — single decision variable
Once fixed, battery power follows: $P_{\text{bat}} = P_{\text{load}} - P_{\text{dc}}$

State: $x(t) \equiv \text{SoC}(t)$
One-dimensional optimal control
 $N = 3600$ decision points

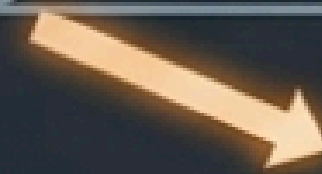
Inertial Force: $m \cdot a$



Rolling Resistance: $m \cdot g \cdot C_r \cdot \cos \alpha$



Aerodynamic Drag: $\frac{1}{2} \rho_{air} C_d A_f v^2$



Grade Force: $m \cdot g \cdot \sin \alpha$

Models — PEMFC Stack (Mann–Amphlett GSSEM)

Three Voltage Loss Mechanisms

Activation V_{act}

Tafel-type; dominant at low current density

Ohmic V_{ohm}

Membrane + contact resistance (Mann correlation)

Concentration V_{con}

Reactant depletion at high current density

$\eta_{\text{fc}} \approx 0.70$ at ~ 12 kW $\rightarrow \approx 0.54$ at peak (128 kW)
Non-linear efficiency makes optimal split non-trivial.

Cell Voltage (GSSEM):

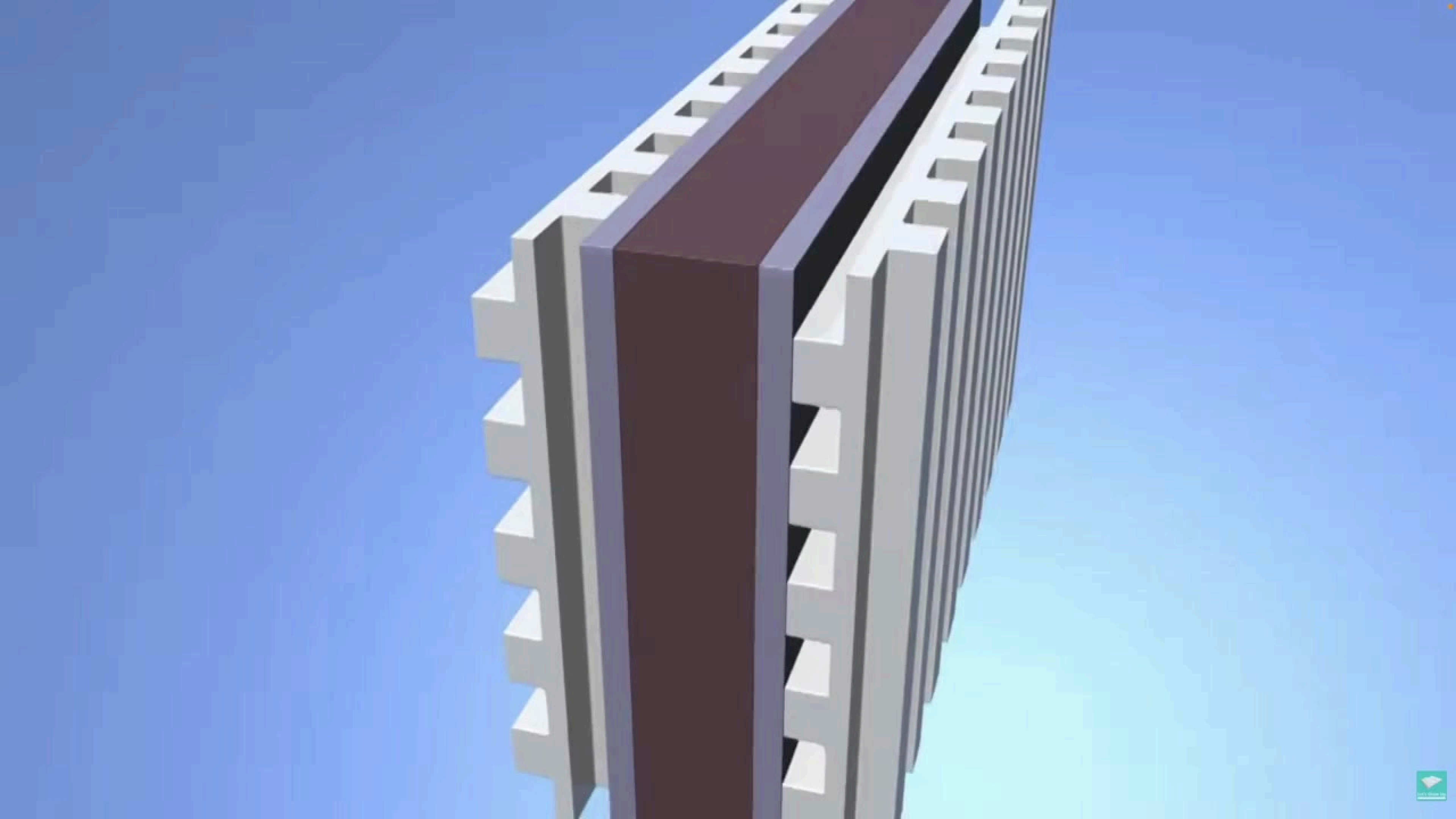
$$V_{\text{FC}} = E_{\text{Nernst}} - V_{\text{act}} - V_{\text{ohm}} - V_{\text{con}}$$

$$N_{\text{cell}} = 330 \cdot A = 300 \text{ cm}^2 \cdot T = 343 \text{ K (isothermal)}$$

H₂ Mass Flow Rate:

$$\dot{m}_{\text{H}_2} = \frac{P_{\text{fc, gross}}}{\eta_{\text{fc}}(P_{\text{fc, gross}}) \cdot \text{LHV}_{\text{H}_2}}$$

$$\text{LHV}_{\text{H}_2} = 120 \text{ MJ/kg} \mid \eta_{\text{fc}} \text{ stored as 1-D lookup table}$$



Models — Battery (1-RC + Hysteresis)

Why 1-RC + Hysteresis?

- LiFePO₄ OCV curve nearly flat between 20%–80% SoC
- Standard RC model cannot distinguish SoC in flat region
- Hysteresis state tracks charge/discharge reference curves
- Nejad et al. 2016: 1.39% mean error — lowest among 10 models

$$\frac{dV_{OC}}{dSoC} = \frac{dV_{OC}^{ref}}{dSoC} + m_{hyst}(V_{OC}^{ref} - V_{OC})$$

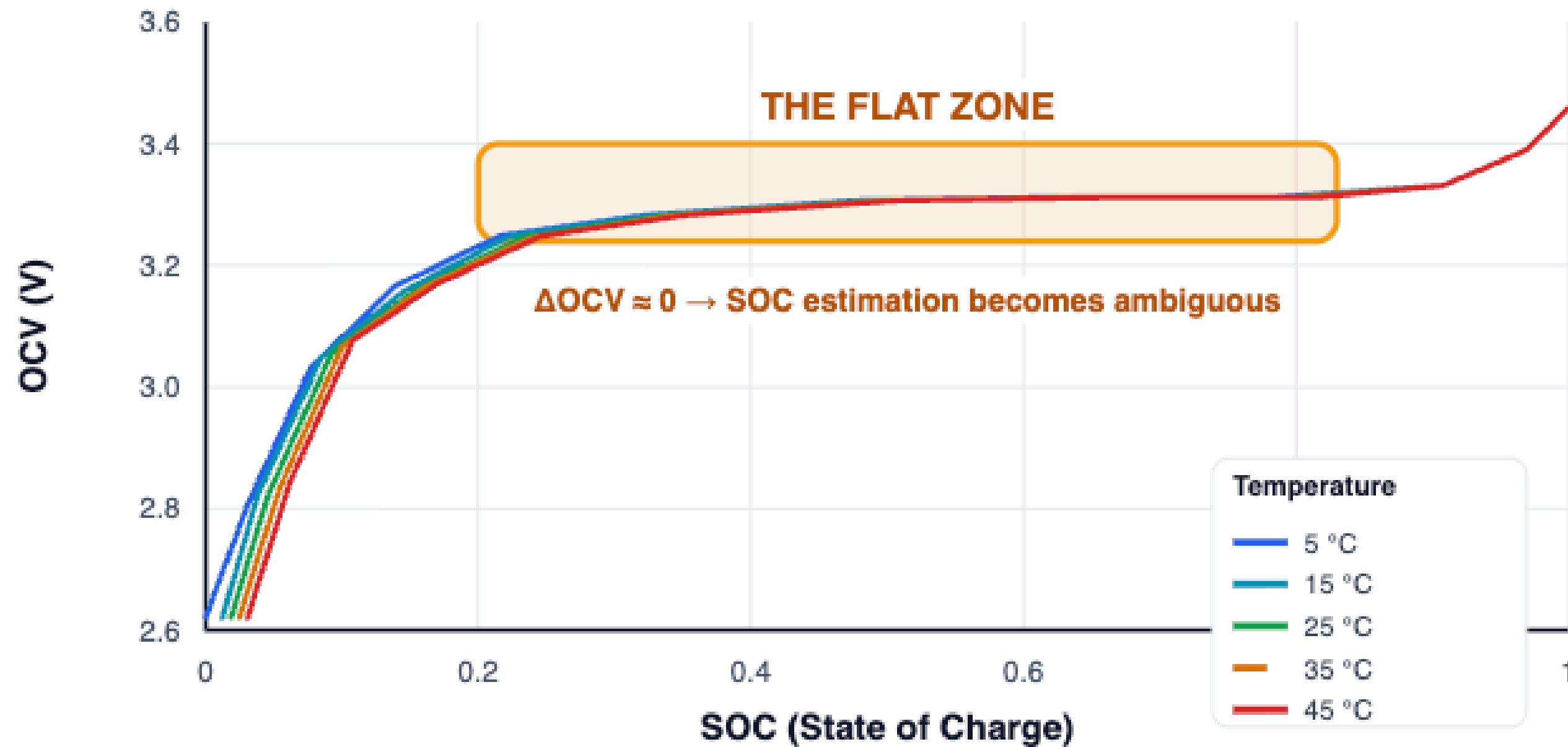
SoC Dynamics (Coulomb Counting):

$$SoC_{k+1} = SoC_k - \frac{\eta_i I_{bat} \Delta t}{Q_{pack}}$$

$\eta_i = 0.995$ (coulombic efficiency)

Battery $\eta \approx 0.87$ vs FC $\eta \approx 0.61 \rightarrow$ Battery wins energetically; but cost decides the optimal split.

The Flat Zone Problem: OCV–SOC Curve



Because the curve is nearly flat here, standard state estimation fails.

Cost Function — The Optimization Objective

Running Cost per Step:

$$L_k = C_{fc,k}(P_{dc,k}) + C_{bat,k}(P_{dc,k}, SoC_k)$$

$C_{fc,k}$ — Hydrogen Cost

$$C_{fc,k} = M_{H_2} \cdot \frac{P_{dc,k}/\eta_{dc} + P_{aux,k}}{\eta_{fc}(\cdot) LHV_{H_2}} \cdot \Delta t$$

$C_{bat,k}$ — Electricity Cost

$$C_{bat,k} = M_{ele} \cdot \frac{P_{bat,k}}{\eta_{eff}(P_{bat,k}, SoC_k)} \cdot \Delta t$$

Total Cost Functional — Principal Mathematical Result :

$$J = \sum_{k=0}^{N-1} L_k(SoC_k, P_{dc,k}) + \gamma (SoC_N - SoC_{target})^2$$

$\gamma = 1000$ €: penalises $SoC_N \neq 0.25$ at magnitude far exceeding expected trip cost ($\approx 5-15$ €).

Four Energy Management Strategies

DP

Dynamic
Programming

Global Benchmark

Bellman backward recursion over full drive cycle. Offline only — cannot be deployed in real vehicle. Serves as theoretical upper bound for all other strategies.

Real-time: No

Comp. load: High (offline)

Optimality: Global

A-ECMS

Adaptive
ECMS

Production Candidate

PMP-based Hamiltonian minimisation at each step. PI feedback on SoC error adapts equivalence factor $s(t)$ online. Near-DP performance in real time.

Real-time: Yes

Comp. load: Low

Optimality: Near-optimal (PMP)

MPC

Model Predictive
Control

Production Candidate

Receding horizon $N_p = 15$. Solves QP at every sample, applies only first step. Assumes constant future load over prediction horizon.

Real-time: Yes

Comp. load: Medium

Optimality: Local

BLFS

Boundary Layer
Surface Following

Protection Layer

Hysteresis band ± 0.05 around SoC_{ref} . FC ramp ≤ 5 kW/s. Lightweight layer operating beneath A-ECMS or MPC — not a standalone strategy.

Real-time: Yes

Comp. load: Very low

Optimality: Rule-based

EMS Formulations — DP & Adaptive ECMS

Dynamic Programming — Bellman Backward Recursion

$$V_k(s) = \min_u \{L_k(s, u) + V_{k+1}(s_{k+1})\}, \quad V_N = \Phi(\text{SoC}_N)$$

State grid: $\Delta \text{SoC} = 0.005$ · Control grid: $\Delta P_{dc} = 1 \text{ kW}$ · $N = 3600$ steps (Sundström & Guzzella 2009; Bertsekas 2005)

Adaptive ECMS — Pontryagin's Minimum Principle

Hamiltonian minimised at each step:

$$H(\text{SoC}, P_{dc}, \lambda) = L(\text{SoC}, P_{dc}) + \lambda \dot{\text{SoC}}$$

Equivalence factor $s(t)$ updated by PI feedback on SoC tracking error:

$$s(t) = s_0 + m e_{\text{SoC}}(t) + n \int_0^t e_{\text{SoC}}(\tau) d\tau$$

EMS Formulations — MPC & BLFS Protection Layer

Model Predictive Control — Receding Horizon ($N_p = 15$)

At every step k , solve a finite-horizon QP and apply only the first control action:

$$\min \sum_{j=k}^{k+N_p-1} L_j(\text{SoC}_j, P_{\text{dc},j}) + \gamma_{\text{MPC}}(\text{SoC}_{k+N_p} - \text{SoC}_{\text{ref}})^2$$

Assumes constant future load over horizon · Solves QP at every sample · Applies first element only

Boundary Layer Surface Following (BLFS) — Protection Layer

SoC > SoC_ref + 0.05 → FC at minimum power → battery depletes toward target

SoC < SoC_ref - 0.05 → FC at peak-efficiency point → battery recharges

SoC inside band $[\pm 0.05]$ → Upper-level (A-ECMS or MPC) command passed through

FC ramp rate limited to $|dP_{\text{fc}}/dt| \leq 5 \text{ kW/s}$ — prevents membrane degradation

Current Status & MAT 4902E Roadmap

MAT 4901E — Completed

- Literature survey (EMS · PEMFC · Battery)
- System architecture fixed (T4 · Mirai II · WLTP)
- Longitudinal vehicle dynamics model
- Mann–Amphlett GSSEM (PEMFC model)
- 1-RC + Hysteresis battery model
- Power flow model & conversion chains
- Cost functional (Eq. 24) derived
- Optimal control problem stated (Eqs. 25–26)
- All 4 EMS strategies formulated (Eqs. 27–31)

MAT 4902E — Upcoming

- 1 Python simulation environment (2-layer)
- 2 PSO/GA parameter identification (Mirai II stack)
- 3 FC efficiency map (η_{fc} lookup table)
- 4 DP implementation (Bellman recursion)
- 5 A-ECMS with PI-adapted equivalence factor
- 6 MPC transcribed as quadratic program
- 7 BLFS integrated beneath A-ECMS & MPC
- 8 Comparison: cost (€/100 km) · H₂ · electricity
- 9 Sensitivity analysis (price scenarios + γ)

The complete mathematical formulation is finalized — MAT 4902E is dedicated to numerical implementation & strategy comparison.

**Thank you
Questions?**